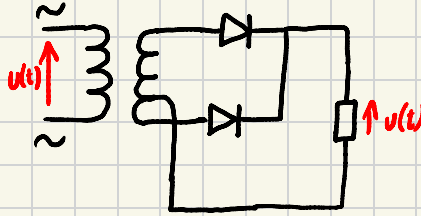
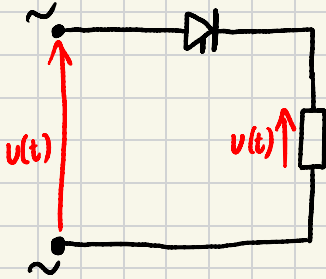
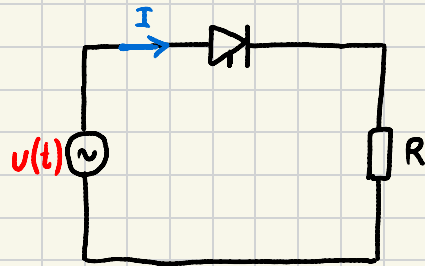


ĆWICZENIA 3

Prostownica sterowalna



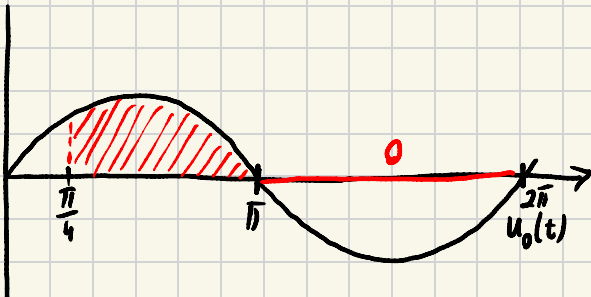
Zad 1



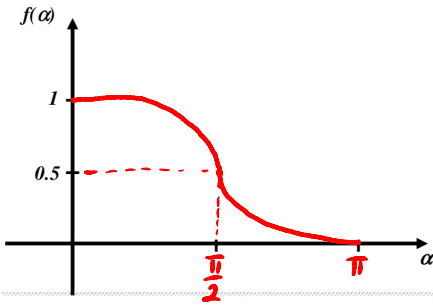
$$\alpha = \frac{\pi}{4}$$

$$U_{sR} = \frac{1}{2\pi} \int_{\alpha}^{\pi} u(t) dt = \frac{1}{2\pi} \int_{\frac{\pi}{4}}^{\pi} 230\sqrt{2} \sin \omega t dt =$$

$$= \frac{230\sqrt{2}}{2\pi} (-\cos \omega t) \Big|_{\frac{\pi}{4}}^{\pi} = \frac{230\sqrt{2}}{2\pi} (-\cos \pi + \cos \frac{\pi}{4}) = \frac{230\sqrt{2}}{2\pi} (1 + \frac{\sqrt{2}}{2}) \approx 88,42 \text{ [V]}$$



Zadanie 2. Proszę narysować charakterystykę sterowania $f(\alpha) = \frac{U_{sr\alpha}}{U_{sr0}}$ dla prostownika z zadania 1 – jednopulsowego.



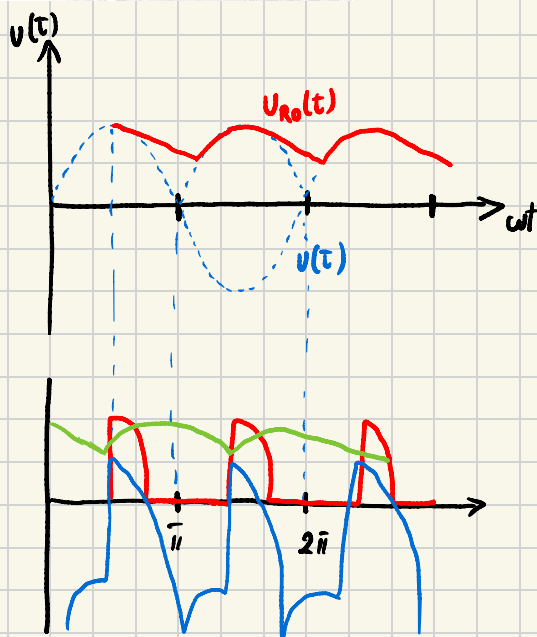
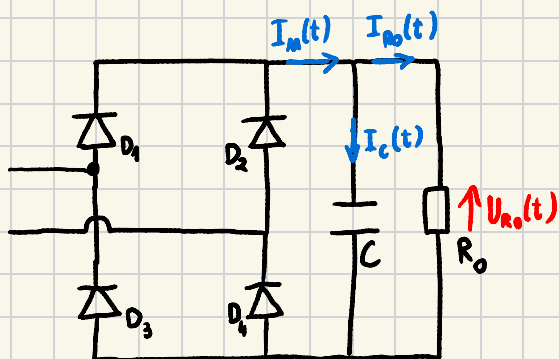
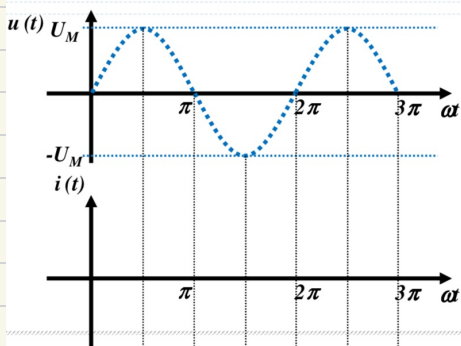
$$f(\alpha) = \frac{U_{sr\alpha}}{U_{sr0}}$$

$$U_{sr0} = \int_0^{\pi} u_m \sin \omega t \, d\omega t = \frac{U_m}{2\pi} (-\cos \omega t) \Big|_0^{\pi} = \frac{U_m}{\pi}$$

$$U_{sr\alpha} = \int_{\alpha}^{\pi} u_m \sin \omega t \, d\omega t = \frac{U_m}{2\pi} (-\cos \omega t) \Big|_{\alpha}^{\pi} = \frac{U_m}{2\pi} (1 + \cos \alpha) = \frac{U_m}{2\pi} \left(\frac{1}{2} + \frac{\cos \alpha}{2} \right)$$

$$f(\alpha) = \frac{\frac{U_m}{2\pi} \left(\frac{1}{2} + \frac{1}{2} \cos \alpha \right)}{\frac{U_m}{2\pi}} = \frac{1}{2} + \frac{1}{2} \cos \alpha$$

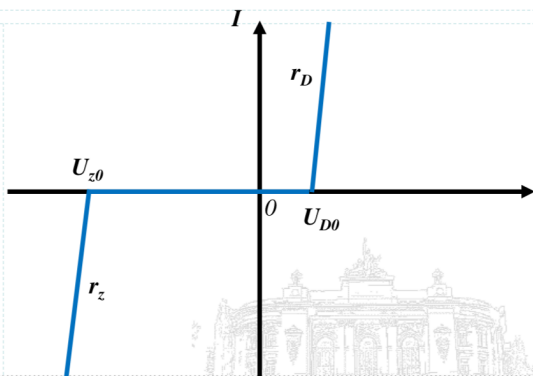
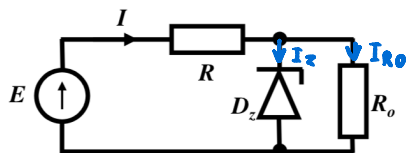
Zadanie 3. Proszę narysować prostownik dwupulsowy, mostkowy, niesterowany, obciążony rezystancyjnie z filtrem pojemnościowym. Proszę narysować przebiegi: $u_s(t)$ – napięcie sieci zasilającej, $u_o(t)$ – napięcie odbiornika, $i_M(t)$ – prąd mostka, $i_c(t)$ – prąd kondensatora, $i_o(t)$ – prąd obciążenia, $i_s(t)$ – prąd źródła.



Zadanie 4. W układzie stabilizatora parametrycznego proszę wyznaczyć zakres zmian rezystancji R_o zapewniający poprawną pracę układu.

Dane: $E=20\text{ V}$, $R=100\ \Omega$, $U_{z0}=4\text{ V}$, $P_z=400\text{ mW}$, $r_z=5\ \Omega$;

Dioda zenera



$$1^\circ R_{0\text{MAX}} = ?$$

$$E = U_R + U_z = (I_{R0} + I_z) \cdot R + U_{z0} + I_{0\text{MAX}} \cdot r_z$$

$$I_{R0} = \frac{U_{R0}}{R_{0\text{MAX}}} = \frac{U_{z0} + I_{0\text{MAX}} \cdot r_z}{R_{0\text{MAX}}}$$

$$E = \left(\frac{U_{z0} + I_{0\text{MAX}} \cdot r_z}{R_{0\text{MAX}}} + I_{0\text{MAX}} \right) \cdot R + U_{z0} + I_{0\text{MAX}} \cdot r_z$$

$$R_{0\text{MAX}} = \frac{(U_{z0} + I_{0\text{MAX}} \cdot r_z) \cdot R}{E - U_{z0} - I_{0\text{MAX}} \cdot (r_z + R)} = \frac{(4 + \frac{94}{4} \cdot 5) \cdot 100}{20 - 4 - \frac{94}{4} (5 + 100)} = 81,8 [\Omega]$$

$$2^\circ R_{0\text{MIN}} = ?$$

$$E = U_R + U_{z0} = I R + U_{z0} = \frac{U_{z0} \cdot R}{R_{0\text{MIN}}} + U_{z0}$$

$$R_{0\text{MIN}} = \frac{U_{z0} \cdot R}{E - U_{z0}} = \frac{4 \cdot 100}{20 - 4} = 25 [\Omega]$$